

Quantum numbers

Quantum numbers are a set of values that describes the state of an electron including its distance from the nucleus, the orientation and type of orbital where it is likely to be found and its spin.

a) Principal quantum number:

The principal quantum number describes the number of the electron's shell/ energy level. It is denoted by n . It can have any positive value; 1, 2, 3 & so on. The smaller n is the lower the energy.

Value of n	1	2	3	4.....
Designation of shells	K	L	M	N.....
	1 st shell	2 nd shell	3 rd shell	4 th shell

n also gives the maximum number of electrons that a shell can hold. A shell with principal quantum number, n can have $2n^2$ electrons. Thus

Maximum number of electrons in K-shell(1 st shell) with $n=1$	$= 2 \times 1^2 = 2$
Maximum number of electrons in L-shell(2 nd shell) with $n=2$	$= 2 \times 2^2 = 8$
Maximum number of electrons in M-shell(3 rd shell) with $n=3$	$= 2 \times 3^2 = 18$
Maximum number of electrons in N-shell (4 th shell) with $n=4$	$= 2 \times 4^2 = 32$

b) Angular momentum / Azimuthal / Subsidiary quantum number:

The subsidiary quantum no. describes the shape of the electron's orbit. The main energy level of electron may be considered as being made up of one or more sub-shells. The no. of sub-shells is limited. It can have any integer value from 0 to $(n-1)$. It is denoted by l .

n	l	Sub-shells
1	0	s
2	0	s
	1	p
3	0	s
	1	p
	2	d
4	0	s
	1	p
	2	d
	3	f

c) Magnetic quantum number:

The third quantum no. was introduced to explain the orientation of electronic orbit in space particularly under the influence of an applied magnetic field. This is known as the magnetic quantum no. and is denoted by m . The allowed values are the integers from $-l$ to $+l$.

n	<i>l</i>	m
1	0	0
2	0	0
	1	-1,0,+1
3	0	0
	1	-1,0,+1
	2	-2,-1,0,+1,+2
4	0	0
	1	-1,0,+1
	2	-2,-1,0,+1,+2
	3	-3,-2,-1,0,+1,+2,+3

d) Spin quantum number:

The spin quantum number represents the direction of the electron spin and is denoted by m_s .

$$m_s = +1/2 \text{ or } -1/2$$

At last we can say that

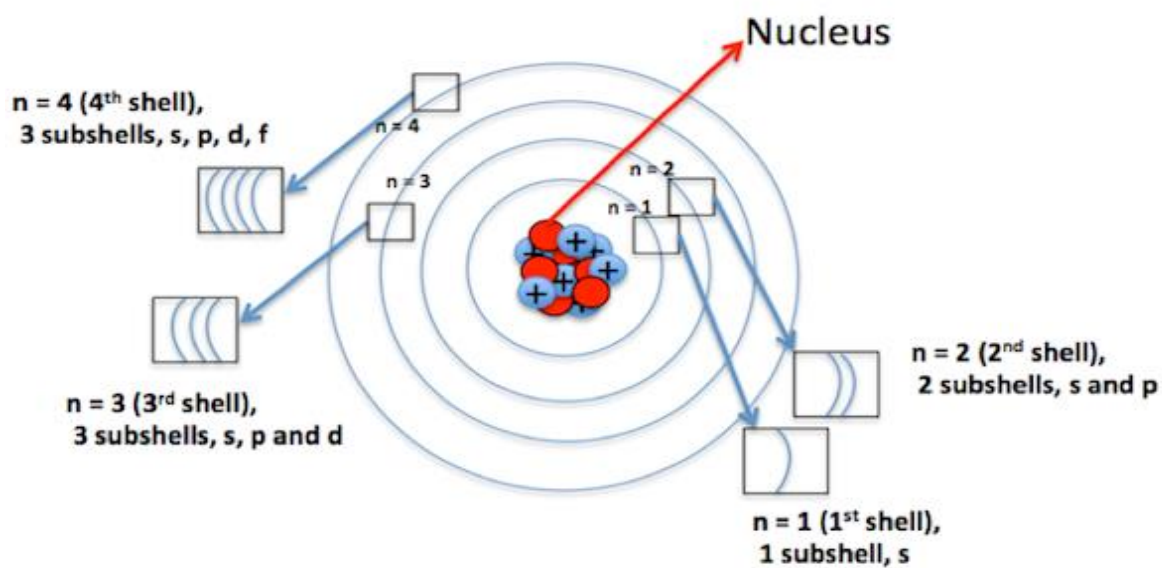
n - describes the energy level

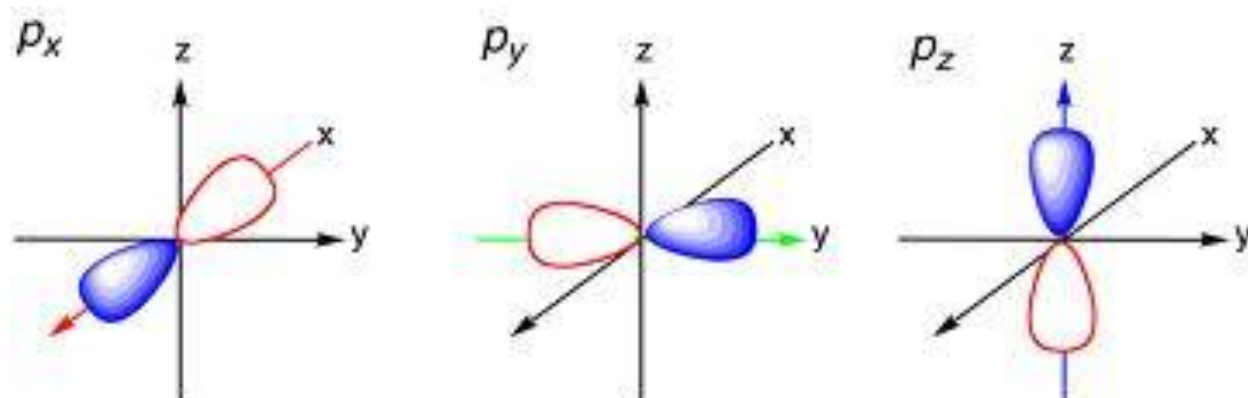
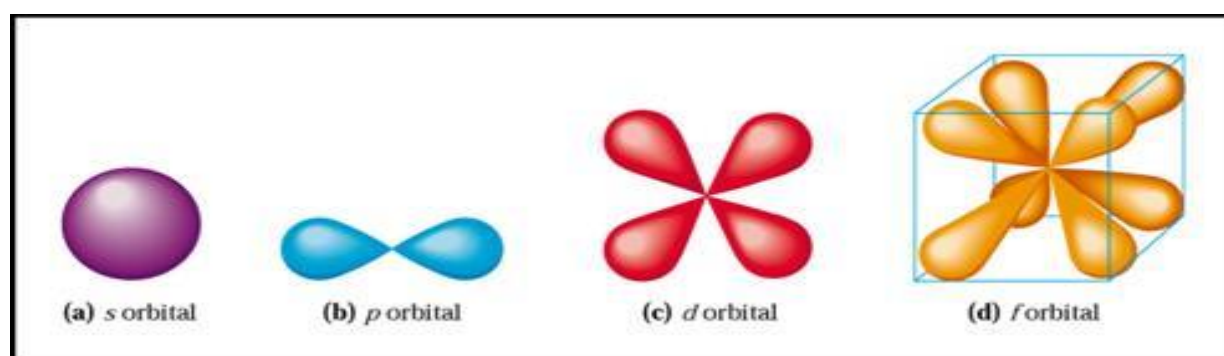
l - describes the subshell

m - describes the orbital of the sub-shell

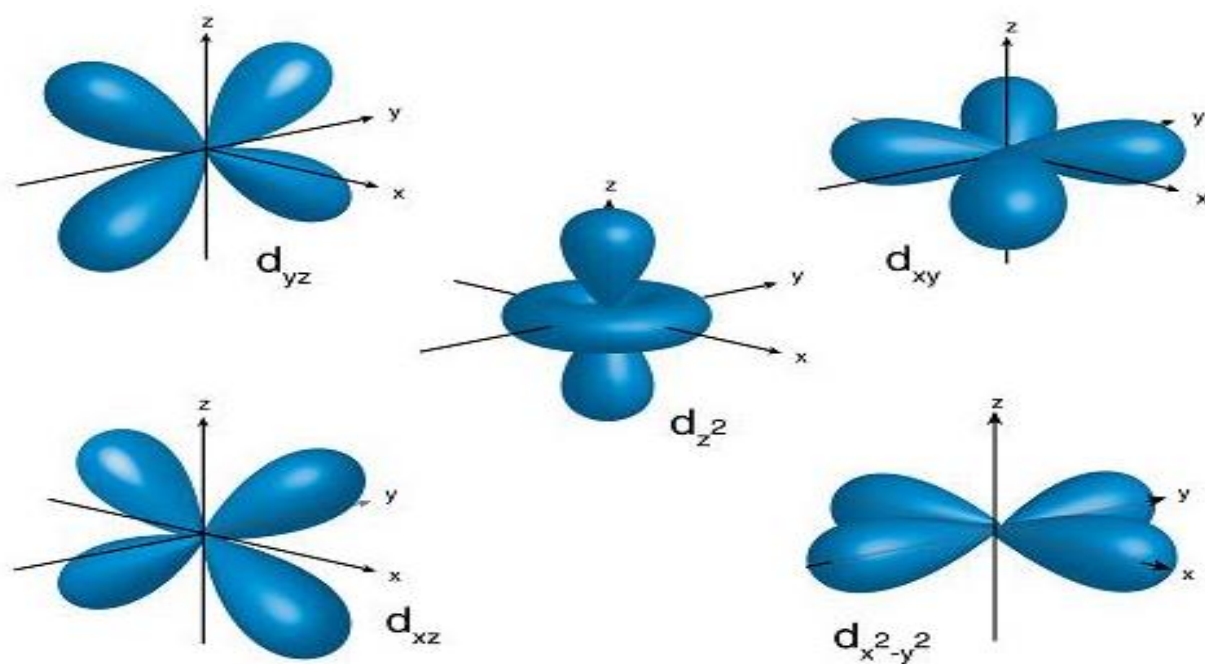
m_s - describes the spin

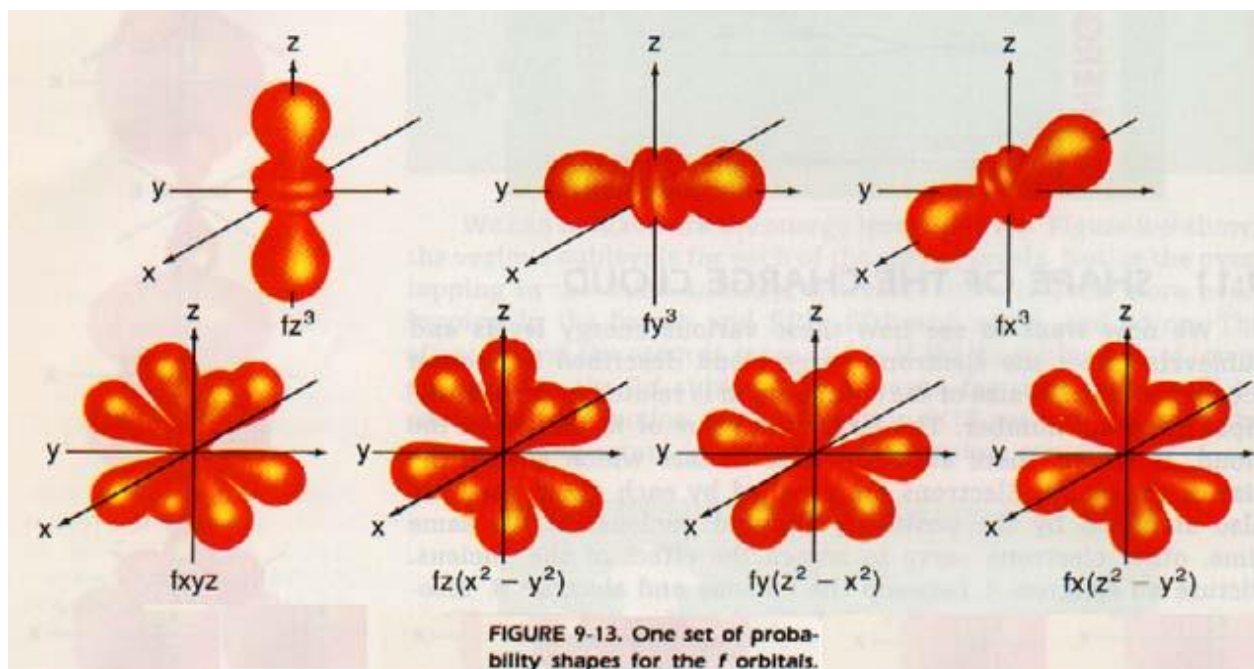
Sub-shells	Orbitals	Maximum no. of electrons
s	1 (s)	2
p	3(p_x, p_y, p_z)	6
d	5	10
f	7	14





The three *p* orbitals are aligned along perpendicular axes





Orbitals and Quantum numbers Practice Questions

1. Explain why each of the following sets of quantum numbers would not be permissible for an electron according to the rules for quantum numbers.

	n	l	m	m_s
i)	1	1	0	$+1/2$
ii)	2	1	0	$-1/2$
iii)	2	0	0	1

2. Which of the following orbitals are not possible? 2s, 2p, 2d & 3f. Explain.
Or, which orbitals cannot exist? 2p 3p 4d 3f 6s 2d. Explain.
3. With the help of the electronic configuration of the elements Cr (24) & Sc (21), deduce the possible values of all the four quantum numbers for the 19th electron in Cr & 21st electron in Sc.
4. What are the shapes of s, p, and d orbitals respectively?
5. How many 1s orbitals are there in an atom? 4p orbitals? 4d orbitals?
6. What is the maximum number of orbitals with:

question part	
$n = 4$	$l = 1$
$n = 2$	$l = 2$
$n = 3$	$l = 2$
$n = 5$	$l = 1 \quad m = -1$

7. Write a set of quantum numbers for a 4f orbital?

8. Describe the electrons defined by the following quantum numbers:

n	<i>l</i>	m
3	0	0
2	1	1
4	2	1
3	3	2
3	1	2

9. If the n quantum no. of an atomic orbital is 4, what are the possible values of *l*? If the *l* quantum no. is 3, what are the possible values of m?

10. Write correct orbital notation for each of the following set of quantum numbers:

	n	<i>l</i>
i)	1	0
ii)	2	1
iii)	3	2
iv)	4	3

11. Find the maximum number of electrons.

- a) For n=5 { Maximum no. of electron in energy level/ main shell = $2n^2$
- b) For all 4d orbitals Maximum no. of electrons in orbital = $2(2l + 1)$ }
- c) For all 5f orbitals
- d) For all n=3 and *l*=1.

12. For n=4 principal shell. Find the following:-

- a) Name of the shell
- b) No. of sub-shells in it
- c) Total no. of orbital's in each sub-shell
- d) No. of orbital's in the principal shell.

13. How many sub-shells are there in the M shell? How many orbitals are there in the f-sub-shell?

14. State the four quantum numbers, then explain the possible values they may have and what they actually represent.

15. State the number of possible electrons described by the following quantum numbers

a) n=3, <i>l</i> = 0
b) n = 3, <i>l</i> = 1
c) n = 3, <i>l</i> = 2, m = -1
d) n = 5, <i>l</i> = 0, m = -2, <i>m_s</i> = -1/2

16. Give the n and *l* values for the following orbitals

- a. 1s b. 3s c. 2p d. 4d e. 5f

17. What are the m values for the following types of orbitals?

- a. s b. p c. d d. f

18. How many possible orbitals are there for n =

- a. 4 b. 5

Pauli Exclusion Principle:

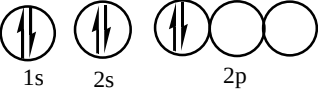
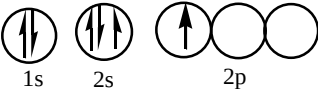
No two electrons in the same atom can have the same value for the four quantum numbers.

e.g. Helium contain two electrons, $1s^2$

	n	<i>l</i>	m	m_s
e_1	1	0	0	+ 1/2
e_2	1	0	0	-1/2

Question:

Which of the following orbital diagrams or electron configurations are possible & which are impossible, according to the Pauli Exclusion Principle? Explain. For those that are allowed, write the electronic configuration.

- i) 
1s 2s 2p
- ii) 
1s 2s 2p
- iii) 
1s 2s 2p

Electronic Configuration

The distribution of electrons in various shells, sub-shells and orbitals of an atom is called its electronic configuration.

		Orbitals			
		s	p	d	f
Principle Quantum Number (Energy Level, "n")	1	1s			
	2	2s	2p		
	3	3s	3p	3d	
	4	4s	4p	4d	4f
	5	5s	5p	5d	5f
	6	6s	6p	6d	6f
	7	7s	7p	7d	7f

Order: 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p

(n + l) rule:

The relative order of energy of different orbitals can also be determined with the help of $(n + l)$ value for a given orbital. This rule is called $(n + l)$ rule, according to which the orbital having the lowest value of $n + l$ has the lowest energy and hence is filled up first with electrons.

When two or more orbitals have the same value of $n + l$, the orbital with lower value of n is lower in energy and hence is filled up first with electrons. Thus the order of energies of; for example, 3d, 4p and 5s orbitals for all of which $n + l = 5$ is as:

$3d(n=3) < 4p(n=4) < 5s(n=5)$; since n for these orbitals is 3, 4 and 5 respectively. Thus orbitals will be filled up with electrons in the following sequence:

$3d, 4p, 5s$
 $\xrightarrow{\hspace{1.5cm}}$
 Orbitals being filled

Rule of half-filled or completely-filled orbitals:

This rule states that the orbitals tend to be half-filled or completely filled, since such orbitals are more stable than other orbitals and hence give more stable configuration to the atom. This rule can be illustrated by considering the electronic configuration of the following elements:

Element(Z= atomic number)	Expected Configuration	Actual(observed) Configuration	Explanation of observed configuration
Cr(24)	$[Ar]4s^23d^4$	$[Ar]4s^13d^5$	One electron from 4s orbital gets shifted to 3d orbital to make 3d orbital half filled $3d^5$ which is more stable than $3d^4$ orbitals.
Cu(29)	$[Ar] 4s^23d^9$	$[Ar] 4s^13d^{10}$	
Mo(42)	$[Kr] 5s^2 4d^4$	$[Kr] 5s^1 4d^5$	
Ag(47)	$[Kr] 5s^2 4d^9$	$[Kr] 5s^1 4d^{10}$	
Pd(46)	$[Kr] 5s^2 4d^8$	$[Kr] 5s^0 4d^{10}$	

Exceptions to Electron Configuration Trends

Period 4:	Period 5:
Chromium : Z:24 $[Ar] 3d^54s^1$	Niobium: Z:41 $[Kr] 5s^1 4d^4$
Copper : Z:29 $[Ar] 3d^{10}4s^1$	Molybdenum: Z:42 $[Kr] 5s^1 4d^5$
	Ruthenium: Z:44 $[Kr] 5s^1 4d^7$
	Rhodium: Z:45 $[Kr] 5s^1 4d^8$
	Palladium: Z:46 $[Kr] 4d^{10}$
	Silver : Z:47 $[Kr] 5s^1 4d^{10}$

Period 6:	Period 7:
Lanthanum : Z:57 $[Xe] 6s^2 5d^1$	Actinium : Z:89 $[Rn] 7s^2 6d^1$
Cerium: Z:58 $[Xe] 6s^2 4f^1 5d^1$	Thorium: Z:90 $[Rn] 7s^2 6d^2$
Gadolinium: Z:64 $[Xe] 6s^2 4f^7 5d^1$	Protactinium: Z:91 $[Rn] 7s^2 5f^2 6d^1$
Platinum: Z:78 $[Xe] 6s^1 4f^{14} 5d^9$	Uranium: Z:92 $[Rn] 7s^2 5f^3 6d^1$
Gold : Z:79 $[Xe] 6s^1 4f^{14} 5d^{10}$	Neptunium: Z:93 $[Rn] 7s^2 5f^4 6d^1$
	Curium: Z:96 $[Rn] 7s^2 5f^7 6d^1$
	Lawrencium: Z:103 $[Rn] 7s^2 5f^{14} 7p^1$

Mendeleev's Periodic Law:

The properties of elements are a periodic function of their **atomic weights**. i.e. if the elements are arranged in the increasing order of their **atomic weights**, the properties of the elements(i.e. similar elements) are repeated after definite regular intervals or periods.

Modern Periodic Law:

The properties of elements are a periodic function of their **atomic numbers**. i.e. if the elements are arranged in the increasing order of their **atomic numbers**, the properties of the elements(i.e. similar elements) are repeated after definite regular intervals or periods.

PERIODIC TABLE

Atomic Properties of the Elements

NIST
National Institute of
Standards and Technology
U.S. Department of Commerce

Frequently used fundamental physical constants

For the most accurate values of these and other constants, visit physics.nist.gov/constants

1 second = 9 192 631 770 periods of radiation corresponding to the transition between the two hyperfine levels of the ground state of ¹³³Cs

speed of light in vacuum c 299 792 458 m s⁻¹ (exact)

Planck constant h 6.6261 x 10⁻³⁴ J s ($h = h/2\pi$)

elementary charge e 1.6022 x 10⁻¹⁹ C

electron mass m_e 9.1094 x 10⁻³¹ kg

$m_e c^2$ 0.5110 MeV

proton mass m_p 1.6726 x 10⁻²⁷ kg

α 1/137.036

R_∞ 10 973 732 m⁻¹

$R_\infty c$ 3.289 842 x 10¹⁵ Hz

$R_\infty h c$ 13.6057 eV

Boltzmann constant k 1.3807 x 10⁻²³ J K⁻¹

□ Solids
■ Liquids
■ Gases
■ Artificially Prepared

Physics Laboratory
physics.nist.gov

Standard Reference Data
www.nist.gov/srd

Group	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
IA	IIA	IIIB	IVB	VB	VIB	VII	VIII	VIII	VIII	VIII	IB	IIB	IIIA	IVA	VA	VIA	VIIA	VIIIA
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
1	2	3	4															

Groups: The vertical columns shown in the periodic table are called groups.

Periods: The horizontal rows shown in the periodic table are called periods.

Defects of Long Form of Periodic Table:

Although the long form of the periodic table is superior to Mendeleev's periodic table in many respects, it remains some of the defects as such.

- The problem of position of hydrogen still remains unsolved.
- Like the Mendeleev's table, it fails to accommodate the lanthanides and actinides in the main body of the table.
- The arrangement is unable to reflect the electronic configuration of many elements.

Aufbau Principle:

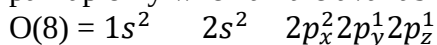
The orbitals are filled up with electrons in the increasing order of their energy.

		Orbitals			
		s	p	d	f
Principle Quantum Number	(Energy Level, "n")	1	2	3	4
		1s	2s	2p	
		3s	3p	3d	
		4s	4p	4d	4f
		5s	5p	5d	5f
		6s	6p	6d	6f
		7s	7p	7d	7f

Order: 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p

Hund's rule of Maximum Multiplicity:

When electrons enter a set of orbitals of a given subshell, electrons will pair up only when all the available orbitals have had at least one electron each.



Q-1: Two elements X and Y have atomic numbers 12 and 16 respectively. Write the electronic configuration for these elements. To which period of the modern periodic table do these two elements belong? What type of bond will be formed between them and why?

Q-2: An element R of atomic number 12 combines separately with NO_3^- , SO_4^{2-} & PO_4^{3-} radicals,

- Write the electronic configuration of element R
- Write the formulae of the three compounds so formed.
- To which group of the periodic table does the element 'R' belong?
- Does it form covalent or ionic compound? Why?

Q-4: A neutral atom of an element has 2K, 8L & 3M electrons. Find out the following from the data:

- Number of protons
- Total no. of principal quantum no.
- Total no. of orbits
- Total no. of P orbital's
- Total no. of sub-shells
- Total no. of p-electrons
- Total no. of paired electrons
- Total no. of unpaired electrons.
- Valency of the elements.

Q-6: Find the number of electrons in atoms which have the following levels in the ground state K shell, L shell, 3s sub-shell & half-filled 3p sub-shell. What is the name of this atom?

Q-7: Which one of the following statements is false for elements given in the periodic table? Explain.

[illegible]

- I.** Electron affinity of Y is larger than others
II. X and T share electrons and form compound.
III. Z has largest atomic radius
IV. Ionic property of ZY compound is larger than others

- Identify the element
- Write the electronic configuration
- Write the balanced equation when it burns in the presence of air and this oxide is dissolved in water.
- Draw the electron dot structure for the formation of this oxide.

Q-9: A element X of group 15 exists as diatomic molecule and combines with hydrogen at 773K in presence of the catalyst to form a compound ammonia which has a characteristic pungent smell.

- Identify the element X, How many valence electrons does it have?
- Draw the electron dot structure of the diatomic molecule of X. What type of bond is formed in it?
- Draw the electron dot structure for ammonia and what type of bond is formed in it?

Q-10: An element X (atomic number 17) reacts with an element Y (atomic number 20) to form a divalent halide.

- Where in the periodic table are elements X and Y placed?
- Classify X and Y as metal, nonmetal or metalloid.
- What will be the nature of element Y? Identify the nature of bonding in the compound formed.
- Draw the electron dot structure of the divalent halide.

Q-11: The outline below is of part of the periodic table. The letters shown represent some of the elements but they are not the symbols of those elements.

[illegible]

Which letters shown on the diagram are:

- (i) Two elements in the same group?
- (ii) Two elements in the same period?
- (iii) Two elements that are metals?
- (iv) An unreactive gas?